HARD · INFORMATION AND IDEAS

Command of Evidence

10

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 INFORMATION AND IDEAS

Early Earth is thought to have been characterized by a stagnant lid tectonic regime, in which the upper lithosphere (the outer rocky layer) was essentially immobile and there was no interaction between the lithosphere and the underlying mantle. Researchers investigated the timing of the transition from a stagnant lid regime to a tectonic plate regime, in which the lithosphere is fractured into dynamic plates that in turn allow lithospheric and mantle material to mix. Examining chemical data from lithospheric and mantle-derived rocks ranging from 285 million to 3.8 billion years old, the researchers dated the transition to 3.2 billion years ago.

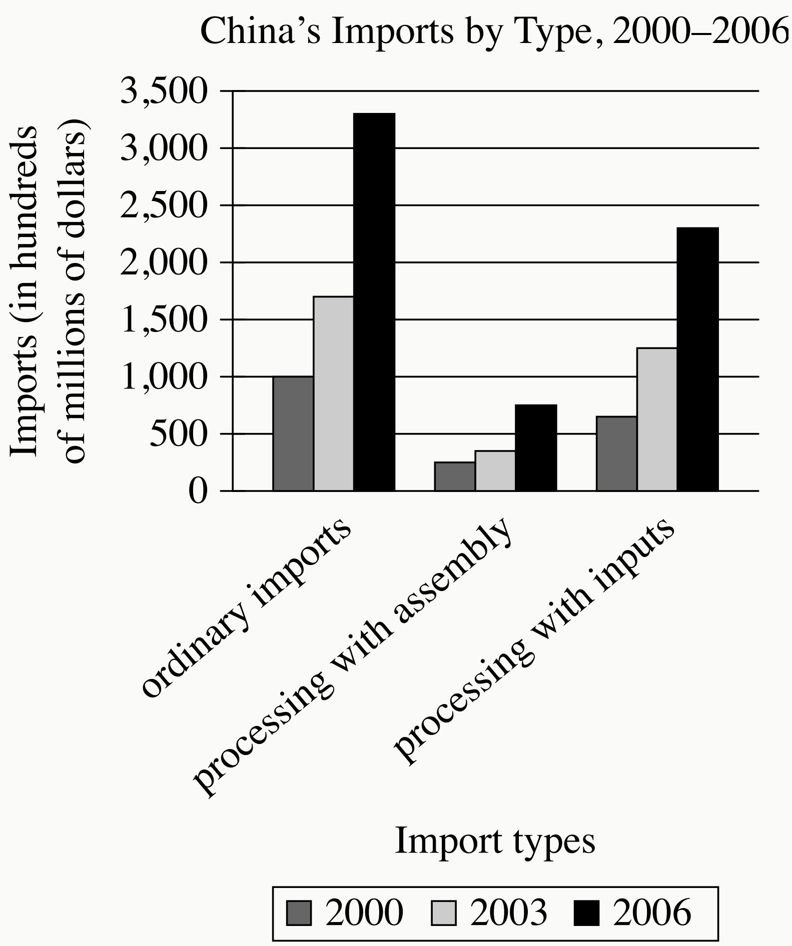
Which finding, if true, would most directly support the researchers' conclusion?

A. Among rocks known to be older than 3.2 billion years, significantly more are mantle derived than lithospheric, but the opposite is true for the rocks younger than 3.2 billion years.

B. Mantle-derived rocks older than 3.2 billion years show significantly more compositional diversity than lithospheric rocks older than 3.2 billion years do.

C. There is a positive correlation between the age of lithospheric rocks and their chemical similarity to mantle-derived rocks, and that correlation increases significantly in strength at around 3.2 billion years old.

D. Mantle-derived rocks younger than 3.2 billion years contain some material that is not found in older mantle-derived rocks but is found in older and contemporaneous lithospheric rocks.



- For each data category, the following bars are shown:
  - 2000
  - 2003
  - 2006
- All values are approximate.
- The Imports data for the 3 categories are as follows:
  - ordinary imports:
    - 2000: 1,000 hundred million dollars
    - 2003: 1,700 hundred million dollars
    - 2006: 3,300 hundred million dollars
  - processing with assembly:
    - 2000: 250 hundred million dollars

        - 2003: 350 hundred million dollars
        - 2006: 750 hundred million dollars
    - processing with inputs:
        - 2000: 650 hundred million dollars
        - 2003: 1,250 hundred million dollars
        - 2006: 2,300 hundred million dollars

A student is researching the Chinese government's 1992 shift to a market economy that emphasizes trade liberalization. One means of trade liberalization involves expanding from ordinary imports into an emphasis on processing imports, which have two types: processing with assembly (in which a firm obtains raw materials from a foreign trading partner without payment and sells the final goods to that partner, charging for assembly) and processing with inputs (in which a firm expends capital to buy raw materials from a trading partner, processes them into final goods, and sells those goods to whichever trading partner it chooses). The student asserts that while initial efforts at trade liberalization were shaped by Chinese firms' limited capital, this situation resolved during the 2000s.

Which choice best describes data from the graph that support the student's assertion?

A. Processing imports with inputs were greater than both ordinary imports and processing imports with assembly in 2006.

B. From 2000 to 2006, processing imports with inputs rose much more sharply than processing imports with assembly did.

C. From 2000 to 2006, neither processing imports with inputs nor processing imports with assembly were greater than ordinary imports.

D. Processing imports with assembly were greater in 2006 than processing imports with inputs in 2000.

Barchester Towers is an 1857 novel by Anthony Trollope. In the novel, Trollope's portrayal of Dr. Proudie underscores the character's exaggerated sense of his own abilities: ______blank

Which quotation from *Barchester Towers* most effectively illustrates the claim?

A. "It must not…be taken as proved that Dr. Proudie was a man of great mental powers, or even of much capacity for business, for such qualities had not been required in him."

B. "[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts, natural and acquired, as must be sure to recommend him to a yet higher notice."

C. "[Dr. Proudie's] residence in the metropolis, rendered necessary by duties thus entrusted to him, his high connexions, and the peculiar talents and nature of the man, recommended him to persons in power."

D. "[Dr. Proudie] was certainly possessed of sufficient tact to answer the purpose for which he was required without making himself troublesome."

Q4 INFORMATION AND IDEAS

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. <u>But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.</u>

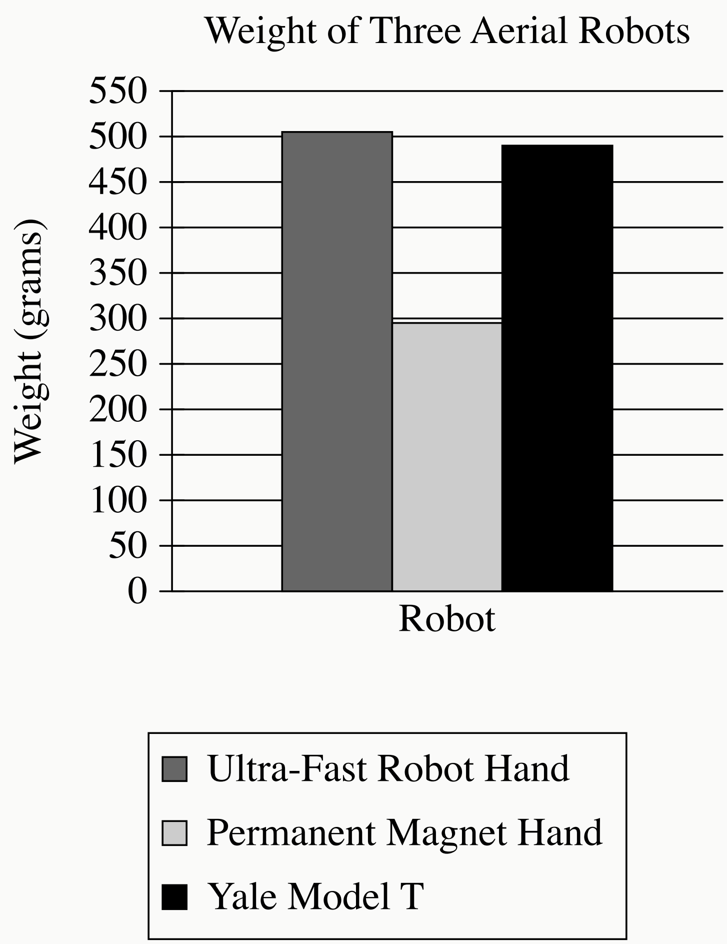
Which finding, if true, would most directly support the claim in the underlined sentence?

A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.

B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.

C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.

D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.



- The data for the 3 categories are as follows:
  - Ultra-Fast Robot Hand: 505 grams
  - Permanent Magnet Hand: 295 grams
  - Yale Model T: 490 grams

Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot's holding force to the robot's weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues' conclusion?

A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.

B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.

C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.

D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

Q6 INFORMATION AND IDEAS

In the “language nest” model of education, Indigenous children learn the language of their people by using it as the medium of instruction and socialization at pre-K or elementary levels. In their 2016 study of a school in an Anishinaabe community in Ontario, Canada, scholars Lindsay Morcom and Stephanie Roy (who are Anishinaabe themselves) found that the model not only imparted fluency in the Anishinaabe language but also enhanced students’ pride in Anishinaabe culture overall. Given these positive effects, Morcom and Roy predict that the model increases the probability that as adults, former students of the school will transmit the language to younger generations in their community.

Which finding, if true, would most strongly support the researchers’ prediction?

A. Anishinaabe adults who didn’t attend the school feel roughly the same degree of cultural pride as the former students of the school feel.

B. After transferring to the school, new students experience an increase in both fluency and academic performance overall.

C. As adults, former students of the school are just as likely to continue living in their community as individuals who didn’t attend the school.

D. As they complete secondary and higher education, former students of the school experience no loss of fluency or cultural pride.

Juvenile Plants Found Growing on Bare Ground and in Patches of Vegetation for Five Species

Species	Bare ground	Patches of vegetation	Total	Percent found in patches of vegetation
T. moroderi	9	13	22	59.1%
T. libanitis	83	120	203	59.1%
H. syriacim	95	106	201	52.7%
H. squamatum	218	321	539	59.6%
H. stoechas	11	12	23	52.2%

Alicia Montesinos-Navarro, Isabelle Storer, and Rocío Perez-Barrales recently examined several plots within a diverse plant community in southeast Spain. The researchers calculated that if individual plants were randomly distributed on this particular landscape, only about 15% would be with other plants in patches of vegetation. They counted the number of juvenile plants of five species growing in patches of vegetation and the number growing alone on bare ground and compared those numbers to what would be expected if the plants were randomly distributed. Based on these results, they claim that plants of these species that grow in close proximity to other plants gain an advantage at an early developmental stage.

Which choice best describes data from the table that support the researchers' claim?

A. For all five species, less than 75% of juvenile plants were growing in patches of vegetation.

B. The species with the greatest number of juvenile plants growing in patches of vegetation was *H. stoechas*.

C. For *T. libanitis* and *T. moroderi*, the percentage of juvenile plants growing in patches of vegetation was less than what would be expected if plants were randomly distributed.

D. For each species, the percentage of juvenile plants growing in patches of vegetation was substantially higher than what would be expected if plants were randomly distributed.

Q8

INFORMATION AND IDEAS

Boldly mixing elements of poetry, fiction, drama, philosophy, and manifesto, Puerto Rican writer Giannina Braschi creates cross-genre literature that explores themes such as immigration and independence. Her works have inspired responses from individuals across different fields and in a wide range of formats, from musical compositions and a comic book to architecture and furniture design. In an essay, a student asserts that the production of these diverse creations by others is reflective of Braschi's own approach to crafting literature.

Which quotation from a scholarly review of Braschi's work best supports the student's claim?

A. "Braschi is the focus of a 2020 collection of essays in which fifteen scholars from seven different countries delved into the linguistic and structural patterns of her writings."

B. "Braschi's eagerness to push boundaries and blend genres within literature invites us to consider how other art forms might also engage with literature."

C. "Before settling in New York City, where she would go on to become a college professor, Braschi studied both literature and philosophy in several cities around the world."

D. "In addition to her creative literary works, Braschi has produced academic pieces analyzing writings by Miguel de Cervantes, Federico García Lorca, and other authors."

Q9

INFORMATION AND IDEAS

A student in a political science course is writing a paper on Aristotle's *The Politics*, in which Aristotle offers his opinion on political instability and gives advice on how constitutions can be preserved. Aristotle observes that different forms of government can fall in different ways—for example, oligarchies might grant power to military leaders during wartime who refuse to relinquish that power during peacetime—but some methods of preserving order apply across all forms of government. The student claims that in particular Aristotle asserts that in a healthy state obedience to law must be as close to absolute as possible and that even minor infractions should not be ignored.

Which quotation from a philosopher's analysis of *The Politics* would best support the student's claim?

A. "When constructing his argument regarding the characteristics of a well-functioning government, Aristotle asserts that 'Transgression creeps in unperceived and at last ruins the state,' illustrating this idea with a comparison to frequent small expenditures slowly and almost imperceptibly chipping away at a fortune until it is ultimately depleted."

B. "When Aristotle writes on the necessity of avoiding corruption in government, he proposes that 'every state should be so administered and so regulated by law that its magistrates cannot possibly make money.' In particular, he thinks oligarchies are particularly susceptible to corruption through bribery."

C. "When Aristotle considers the health of constitutions, he states that 'Constitutions are preserved when their destroyers are at a distance, and sometimes also because they are near, for the fear of them makes the government keep in hand the constitution.' He holds that rulers who wish to see constitutions preserved must continually remind the populace of the dangers that would result from a constitutional collapse."

D. "When contrasting different forms of government, Aristotle holds that 'oligarchies may last, not from any inherent stability in such forms of government, but because the rulers are on good terms both with the unenfranchised and with the governing classes.' That is, oligarchic leaders who wish to hold on to power will introduce members of disenfranchised classes into government in a participatory role."

Q10 INFORMATION AND IDEAS

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep's and goats' diets, while the cattle's diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team's conclusion?

A. Analysis of the animal bones showed that the cattle's diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.

B. Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.

C. Cattle's diets generally require larger amounts of food and a greater variety of nutrients than do sheep's and goats' diets.

D. The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.